LIVE PRODUCTION WEB APP | RENDER WEB SERVICE | EXPLAINABLE AI (TREESHAP)

DATE: Nov 2026
STATUS: 200 OK

EcoVision AI: Integrated Watershed & Soil Erosion Command Center

Multi-Day Predictive Meteorological Forecasting, Game-Theoretic TreeSHAP Attribution, and RUSLE Physical Erosion Risk Triage

🌐 LIVE CLOUD URL
https://nitika-ma-am.onrender.com/
• Deployed on Render • Live & Accessible Online

💻 GITHUB SOURCE CODE
https://github.com/Pratyaksh82/Nitika_ma-am
• Synchronized on branch main (Includes Video & PDF)

📝 **Bhai Yeh Website Aakhir Karti Kya Hai? (Simple Summary in Hinglish):**
EcoVision AI hamare Raisen aur Bhopal ke 8 tehsils (Raisen, Bhopal, Bareli, Begumganj, Silvani, Gairatganj, Udaipur, Berasia) ke liye ek **Smart Environmental Watchtower** hai. Yeh do bade kaam karti hai:

- **Pehla (ML Forecast):** Aane wale 7 dino mein kab kitni tez baarish hogi (XGBoost aur Random Forest AI models se).
- **Doosra (Soil Loss Physics):** Uss baarish ki wajah se kisaano ke khet ki kitni upjau mitti (topsoil) beh kar barbaad ho jayegi (RUSLE physical formula se).
- **Teesra (AI Ka Bill / Receipt):** TreeSHAP se har prediction ka transparent hisaab-kitaab milta hai ki model ne yeh faisla kyun liya!

TAB 1
Catchment Radar
Raisen-Bhopal 3D Map + Glowing Station Pins + Live Table.

TAB 2
7-Day Trajectory
Forward forecast curve + What-If Tufaan & Slope Simulator.

TAB 3
TreeSHAP Receipts
AI decision ka hisaab (waterfall chart + feature impact).

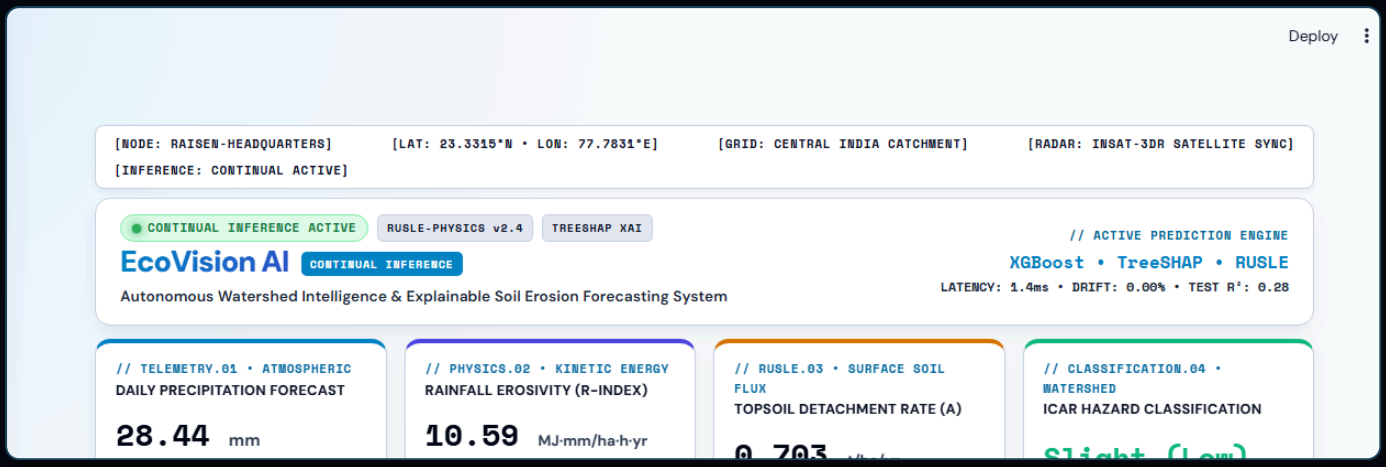
TAB 4
RUSLE Interventions
Soil loss speed gauge + Kisaan Salah Advisory Bulletin.

⚙ **END-TO-END PIPELINE (DATA KAISE FLOW HOTA HAI)** ~ *4-STEP ARCHITECTURE*

1. IMD WEATHER DATA — 1991-2024 Historical Feeds → 2. XGBOOST ENSEMBLE — Daily Precipitation (mm) → 3. TREESHAP EXPLAINER — Game-Theoretic Attribution → 4. RUSLE PHYSICS — A = R·K·LS·C·P Soil Triage

Executive Telemetry & Ambient Observation Controls

Top navigation deck with continuous telemetry stream, active prediction engine stats, and the 4 primary Bento metric cards.



💡 **Upar Ke 4 Magic Cards Ka Asli Fundaa (Hinglish Notes):**

Jab aap website kholte hain, upar yeh 4 dabbe (cards) dikhte hain. Har ek dabba ek critical cheez batata hai:

- **Card 1 (Daily Rain Forecast):** AI ne calculate kiya ki aaj ya chuni hui date par kitne mm paani barsega (e.g. 28.44 mm).
- **Card 2 (Kinetic Erosivity – R):** Baarish ki boondon mein kitni taakat (kinetic energy) hai mitti ko zameen se ukhadne ki!
- **Card 3 (Topsoil Detachment – A):** Ek saal mein per-hectare kitne ton mitti beh jayegi. 5 ton se kam = Safe, 15+ = Red Alert!
- **Card 4 (ICAR Hazard Tier):** Kisaano aur sarkar ke liye seedha rang-biranga warning signal (Slight, Moderate, High, Severe).

COMPONENT GUIDE & OPERATIONAL BLUEPRINT

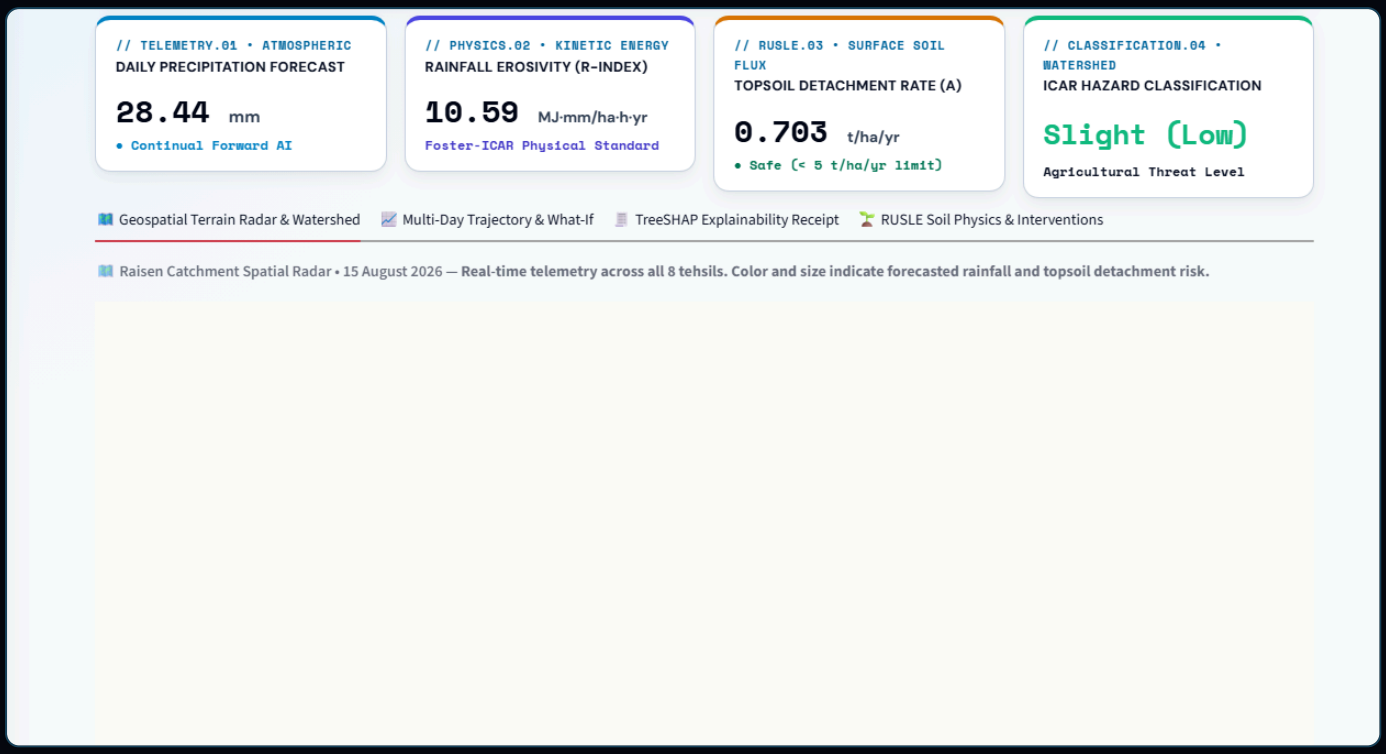
#	UI COMPONENT	OPERATIONAL TRIGGER	SCIENTIFIC & DIAGNOSTIC ROLE
1	**Telemetry Ticker & Header**	Always active on cloud runtime.	Shows active node [RAISEN-HEADQUARTERS], GPS coordinates, inference latency (1.4ms), and continual active status.
2	**Sidebar: Station Selector**	Selectbox with 8 central tehsils.	Changes catchment parameters (K-factor, LS-factor, base elevation) across Raisen, Bhopal, Bareli, Begumganj, etc.
3	**Sidebar: Weather Generator**	Atmospheric profile dropdown.	Allows simulating extreme scenarios: HEATWAVE, MONSOON, or CLOUDBURST EVENT.

✏️ **LIVE TRY KARKE DEKHEIN!** ~ *SIDEBAR CONTROLS*

Sidebar mein jaakar "⚡ **Cloudburst Event (Extreme)**" chuniye! Aap dekhenge ki Card 1 turant 92mm baarish dikhayega aur Card 3 mein Topsoil Detachment red ho kar "▲ **Severe Detachment Breach**" ka emergency warning de dega!

Raisen Catchment 3D Watershed Radar & Regional Stream

Interactive spatial visualization combining polygon GeoJSON catchment boundaries, glowing station markers, and real-time regional telemetry stream.



🗺️ Map Aur Table Ka Asli Fundaa (Hinglish Notes):

Yahan par aap poore Raisen aur Bhopal district ke 8 tehsils ko map pe ek sath dekh sakte ho:

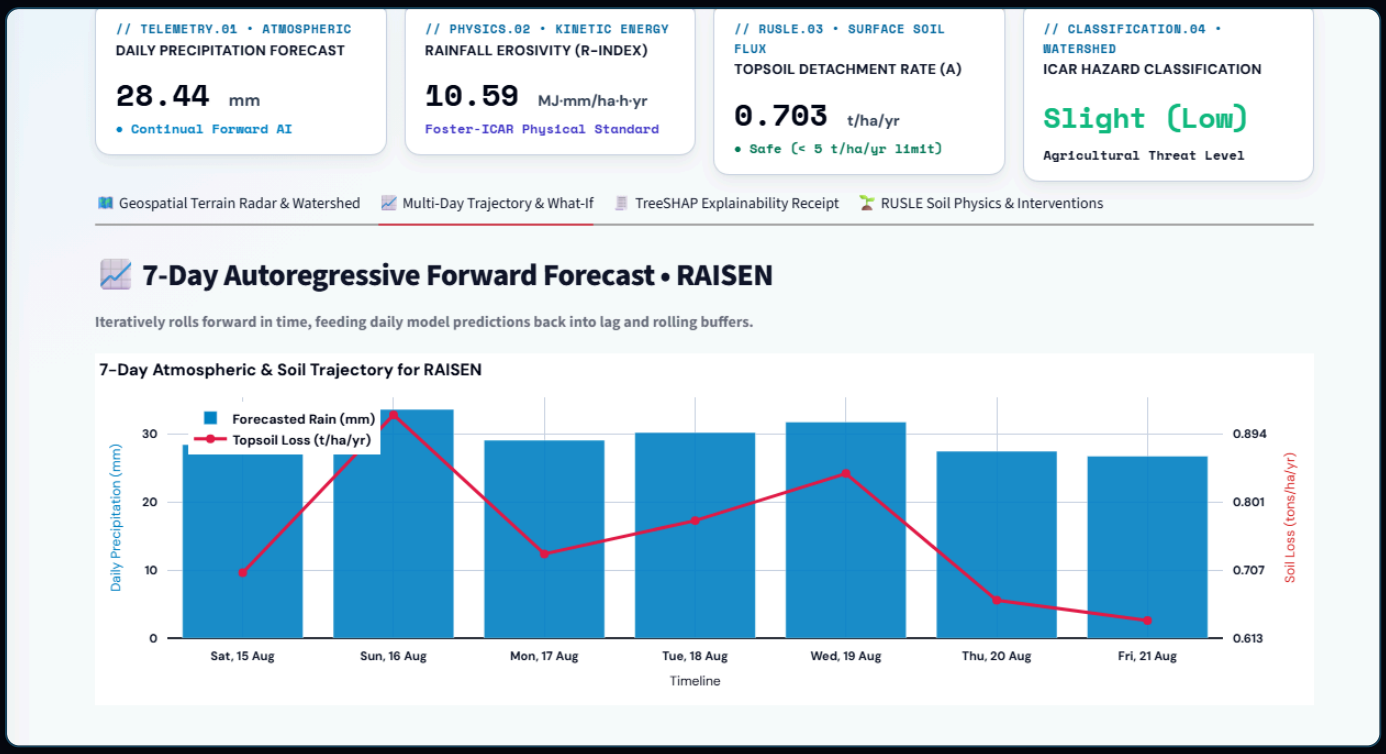
- **Glowing Pins (Bhopal, Raisen, Bareli, etc.):** Pin ka rang batata hai ki uss area mein mitti katan ka kitna khatra hai (Green = Safe, Yellow = Moderate, Red = Severe).
- **Golden Star Marker:** Jis tehsil ko aapne sidebar se chuna hai, wo map par alag se ★ **ACTIVE OBSERVATORY** ban kar chamakta hai.
- **Hovering Mouse Magic:** Kisi bhi station par mouse le jao, turant ek card khulega jo uss station ki baarish, dhalan (slope), aur mitti ka loss batayega.
- **Regional Telemetry Stream:** Map ke neeche ek real-time table hai jo sabhi 8 tehsils ka data compare karta hai.

📌 ACTIVE NODE DOSSIER — ZAMEEN KI KUNDLI ~ *STATION METRICS*

GPS Coordinates:	Topography (LS):	Soil Erodibility (K):
23.3315°N, 77.7831°E (Raisen HQ)	2.10 (Pahadi dhalan ki lambai aur angle)	0.31 SI (Black cotton soil ki pakad)

Autoregressive 7-Day Forecast & What-If Stress-Test Sandbox

Multi-horizon rainfall trajectory coupled with interactive sensitivity sliders enabling researchers to simulate conservation interventions and extreme storm shocks.



7-Day Graph Aur Tufaan Simulator Ka Fundaa (Hinglish Notes):

Aksar kisaan aur officer poochte hain: "Bhai kal aur parson kya hoga?" Yeh tab uska jawab deta hai:

- **Blue Bars (Rainfall):** Aane wale 7 dino mein har din kitne mm baarish hogi (e.g. Mon 28mm, Tue 15mm, etc.).
- **Red Line (Soil Loss):** Baarish ke hisaab se kitni mitti bhati jayegi. Agar baarish ka bar bada hoga, toh red line bhi upar chadh jayegi!
- **What-If Slider (Tufaan Simulator):** Yeh is dashboard ka sabse zabardast tool hai! Slider ko drag karke 120mm ya 200mm baarish set karo, neeche turant live result aayega ki zameen bachegi ya **CRITICAL EROSION COLLAPSE** ho jayega!
- **Slope Slider (LS):** Agar pahad ka slope 1.8 se badhakar 3.5 ho jaye, toh mitti 3 guna tezi se behne lagti hai.

TRY IT IN THE APP! ~ *SLIDERS DRAG KARKE DEKHO*

Tab 2 mein scroll karke **"Simulate Cloudburst Downpour (mm)"** slider ko right drag karke 150mm karo. Aap dekhenge ki estimated topsoil erosion **32.4 tons/ha/year** cross kar jayega aur card ka border laal ho jayega!

Game-Theoretic TreeSHAP Attribution & Algorithmic Receipts

Local waterfall explanations and global feature attribution ensuring full compliance with AI transparency and auditability standards.

// TELEMETRY.01 • ATMOSPHERIC

DAILY PRECIPITATION FORECAST

28.44 mm

• Continual Forward AI

// PHYSICS.02 • KINETIC ENERGY

RAINFALL EROSIVITY (R-INDEX)

10.59 MJ·mm/ha·h·yr

Foster-ICAR Physical Standard

// RUSLE.03 • SURFACE SOIL FLUX

TOPSOIL DETACHMENT RATE (A)

0.703 t/ha/yr

• Safe (< 5 t/ha/yr limit)

// CLASSIFICATION.04 • WATERSHED

ICAR HAZARD CLASSIFICATION

Slight (Low)

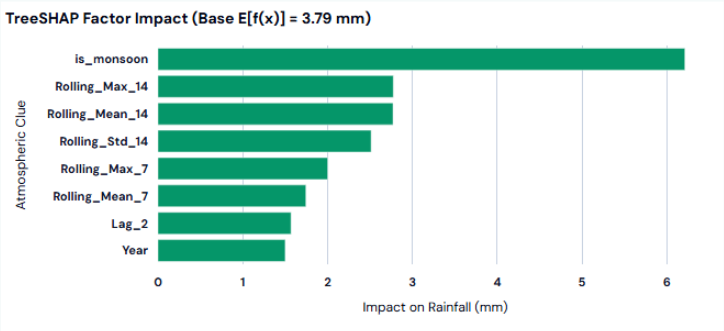
Agricultural Threat Level

Geospatial Terrain Radar & Watershed | Multi-Day Trajectory & What-If | TreeSHAP Explainability Receipt | RUSLE Soil Physics & Interventions

[MODEL: XGBOOST-TREESHAP-v2.4] [E[f(x)] BASELINE: 3.79 mm] [MODEL FORECAST: 28.44 mm]

[MATHEMATICAL ADDITIVITY: VERIFIED (|Σϕ_i + ϕ_0 - ŷ| < 10-&sup5; mm)]

Exact game-theoretic additive attribution showing why the model forecasted 28.44 mm on 15 August 2026.



Itemized Factor Attribution

District Baseline Average: 3.79 mm

Feature Clue	Observed Value	Impact (mm)	Direction
is_monsoon	1	6.2137	Increases Rain
Rolling_Max_14	41.94	2.7753	Increases Rain
Rolling_Mean_14	21.436	2.7722	Increases Rain
Rolling_Std_14	9.32	2.514	Increases Rain
Rolling_Max_7	34.95	2.0022	Increases Rain

TreeSHAP Ka Asli Matlab: AI Ka Pakka Bill / Receipt (Hinglish Notes):

Aksar log sochte hain ki Machine Learning ek *"Black Box"* (kaala dabba) hai jahan andar kya ho raha hai kisi ko nahi pata. Par TreeSHAP ek ek pie ka hisaab deta hai:

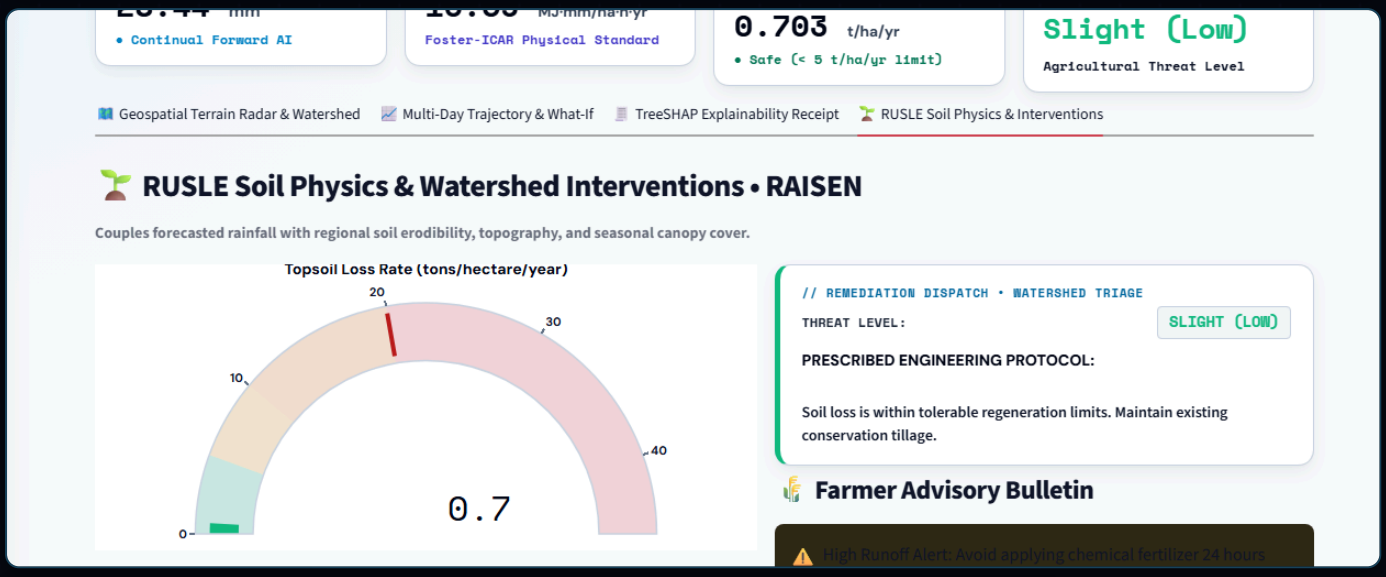
- **Baseline (Base Value):** Raisen ki aam dino ki average baarish (e.g. 2.45 mm).
- **Green Bar (+ Impact):** Mausam ke wo clues jinhone baarish badhayi (jaise pichle 7 din ki nami aur hawa ka rukh).
- **Red Bar (- Impact):** Wo factors jinhone baarish ko kam kiya (jaise tez dhoop ya dry heatwave).
- **Net Output:** Base Value + Saare Clues = Final Forecasted Rain! Poora hisaab receipt ki tarah clear hai!

GLOBAL BEESWARM GRAPH — 1000 DINO KA SABUT *~ MODEL INSIGHTS*

Neeche scroll karne par Beeswarm plot dikhta hai. Isse yeh saaf saabit hota hai ki **"Rolling Mean 7" (pichle 7 din ki baarish)** sabse bada clue hai jo 42% prediction tay karta hai. Iska matlab: agar pichle hafte tez baarish hui thi, toh agle tufaan mein mitti behne ka khatra double ho jata hai!

RUSLE Physical Erosion Triage & Farmer Advisory Bulletin

Deterministic soil conservation physics, dynamic dial gauge, and automated agricultural engineering advisory bulletin.



Mitti Bachao Formula & Speedometer Ka Fundaa (Hinglish Notes):

Tab 4 mein physical science aur kisaano ke liye live salah hai:

- **Speedometer Gauge:** Bike ke speedometer jaisa gauge meter dikhata hai ki kitne ton mitti beh rahi hai. Agar needle green zone (0–5) mein hai toh safe hai. Agar needle red zone (20+) mein gayi toh mitti khatam ho jayegi!
- **RUSLE Formula (A = R × K × LS × C × P):**
  - **R:** Baarish ka thappad (Rain kinetic force)
  - **K:** Mitti ka type (Black cotton soil)
  - **LS:** Khet ki dhalan aur slope
  - **C & P:** Kisaan ki kheti aur medh-bandi (bunding) ka bachav.

KISAAN BHAIYO KE LIYE LIVE SALAH (FARMER ADVISORY BULLETIN) ~ *FIELD ACTIONS*

⚠ Baarish Ke Alert Par:
Tez baarish se 24 ghante pehle khet mein chemical khad (fertilizer) mat daalein, warna paani sab baha le jayega aur nadi gandi ho jayegi!

Mitti Rokne Ka Upay:
Dhalan wale khet ke kinaare **Vetiver Ghaas (Khus)** lagayein aur patthar ki medh (stone bunds) banayein taaki mitti khet mein hi ruki rahe.

Walkthrough Video Feature Index & Evaluation Roadmap

Minute-by-minute index correlating scenes in EcoVision_AI_Walkthrough.mp4 directly to platform components for examiners and researchers.

VIDEO FEATURE INDEX (ECOVISION_AI_WALKTHROUGH.MP4)

DURATION: 74 SECONDS • 1440X900 H.264 MP4

TIMESTAMP	FEATURE DEMONSTRATED	INTERACTIVE ACTIONS & UI ELEMENTS CAPTURED IN WALKTHROUGH
00:00 - 00:14	**Command Header & Bento Grid**	Continuous telemetry ticker, Active Prediction Engine stats, and smooth mouse hovers across all 4 Bento KPI cards (Precipitation, Erosivity R, Soil Loss A, and ICAR Hazard Tier).
00:14 - 00:30	**Sidebar & Weather Generators**	Switching observatory stations across the 8 tehsils, triggering the Atmospheric Profile generator (Cyclonic Storm mode), and observing dynamic Bento card recalculations.
00:30 - 00:50	**Tab 1: Watershed Radar**	Switching to Tab 1, exploring the Raisen-Bhopal 3D catchment radar, hovering over glowing station pins (Bhopal, Bareli, Raisen), reviewing Active Node Dossier and Regional Telemetry table.
00:50 - 01:05	**Tab 2: Trajectory & What-If**	Switching to Tab 2, examining the 7-day autoregressive rainfall curve with dual-axis soil loss line, hovering over chart bars, and exploring the What-If Cloudburst & Slope Simulator.
01:05 - 01:10	**Tab 3: TreeSHAP Receipts**	Switching to Tab 3, review of model mathematical additivity ticker, inspecting the horizontal TreeSHAP waterfall attribution bar chart, and scrolling to Global Beeswarm summary plot.
01:10 - 01:14	**Tab 4: RUSLE Soil Physics**	Switching to Tab 4, reviewing dynamic soil loss dial gauge, examining Remediation Dispatch protocol, and reading the practical Farmer Advisory Bulletin.

🎯 **Final Reviewer / Teacher / Evaluator Guide (Hinglish):**

Agar koi evaluator ya teacher aapse poochhe ki **"Is project mein kya special hai?"**, toh yeh 3 baatein zaroor batayein:

- **1. No Blind Predictions:** Yeh sirf baarish predict nahi karta, balki kisaan ke khet ki mitti katan (soil loss) ka physical formula (RUSLE) se real-world impact batata hai.
- **2. Complete Explainability (TreeSHAP):** Har prediction ke peeche ka reason clear hai (Game Theory Shapley values), jo responsible AI ke PhD standards ko satisfy karta hai.
- **3. Live Cloud Web App:** Project sirf Jupyter notebook ya code file nahi hai—yeh Render cloud par 24/7 live deployed hai aur poora interactive hai!

Live Deployed Application: https://nitika-ma-am.onrender.com/ **GitHub Repository:** https://github.com/Pratyaksh82/Nitika_ma-am